

Lecture 20: Singular Value Decomposition

Balls, Gaussian Clouds, and Real Orthogonal Axes

A Statistical Question

Today's Question

In statistics, a standard normal random vector is **round**:

$$x \sim N(0, I).$$

Every direction has the same variance.

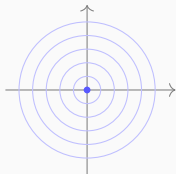
Now apply a real matrix:

$$y = Ax.$$

What shape is the distribution of y ?

Round Cloud Becomes Ellipse Cloud

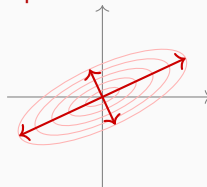
round normal cloud



input space
 $x \sim N(0, I)$



ellipse normal cloud

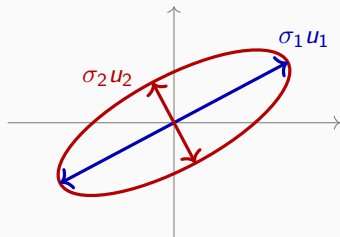


output space
 $y = Ax$

SVD asks for the axes and axis lengths of this ellipse.

What Information Do We Need?

The output cloud has two visible pieces of geometry:



- **Directions:** where are the axes? $u_1, u_2, \dots$
- **Lengths:** how long are the axes? $\sigma_1, \sigma_2, \dots$

Covariance Remembers the Shape

For a random vector, covariance records the shape of the cloud.

If $y = Ax$, then:

$$\text{Cov}(y) = \text{Cov}(Ax) = A \text{Cov}(x) A^T.$$

Linear transformations change covariance by $C \mapsto ACA^T$.

Standard Normal Input

For a standard normal input:

$$\text{Cov}(x) = I.$$

Therefore:

$$\text{Cov}(y) = AIA^T.$$

So:

$$\boxed{\text{Cov}(y) = AA^T.}$$

The First Symmetric Matrix

The output cloud is controlled by the real symmetric matrix

$$AA^T, \quad (AA^T)^T = AA^T.$$

So the previous lecture applies:

$$AA^T = \underbrace{U}_{\text{orthogonal axes}} \underbrace{\Sigma^2}_{\text{variances along axes}} \underbrace{U^T}_{\text{measure in those axes}}.$$

The output axes come from real symmetric diagonalization.

No Unitary Matrix Is Needed Here

This lecture is an applied real-number story.

The visible objects are:

$$A^T, \quad AA^T, \quad A^T A, \quad U^T U = I, \quad V^T V = I.$$

The final real formula will be:

$$A = U \Sigma V^T.$$

Complex notation is only a later remark: replace T by H .

Read the Formula in Three Steps

The formula

$$A = U\Sigma V^T$$

is not just symbols. It is a three-step geometric story.

Step	Matrix	Geometric action
1	V^T	rotate the input axes
2	Σ	stretch by axis lengths
3	U	rotate to the output axes

Σ stores lengths. U stores output directions. V stores input directions.

The Same Question Without Probability

Forget Probability for One Minute

The same shape question can be asked without random variables.

Look at the unit ball:

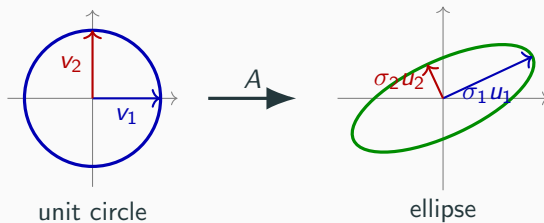
$$B_1 = \{x : \|x\| \leq 1\}.$$

Apply a matrix:

$$A(B_1) = \{Ax : \|x\| \leq 1\}.$$

What shape is $A(B_1)$?

Circle to Ellipse



The image of a round object has special axes.

Axis Equation

The special input directions are:

$$v_1, v_2, \dots$$

They land exactly on the output axes:

$$Av_1 = \sigma_1 u_1, \quad Av_2 = \sigma_2 u_2, \quad \dots$$

input axis	stretch	output axis
v_1	σ_1	u_1
v_2	σ_2	u_2
v_3	0	invisible

Definition (Singular Values)

The singular values of a real matrix A are the axis lengths of the ellipsoid

$$A(B_1) = \{Ax : \|x\| \leq 1\}.$$

They are written:

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

A singular value is a length, so it is never negative.

What We Must Find

SVD asks for three pieces of data:

$$Av_i = \sigma_i u_i$$

Symbol	Meaning	Where it lives
v_i	input axis direction	domain of A
σ_i	axis length	nonnegative number
u_i	output axis direction	codomain of A

Once we know these, the matrix itself becomes easy to describe.

Stack the Axis Equations

Write all axis equations at once:

$$A \underbrace{\begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix}}_V = \underbrace{\begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix}}_U \underbrace{\begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}}_{\Sigma}.$$

Short version:

$$\boxed{AV = U\Sigma.}$$

From Axis Equations to SVD

The input axes are orthonormal, so:

$$V^T V = I, \quad V^{-1} = V^T.$$

Starting from:

$$AV = U\Sigma,$$

multiply on the right by V^T :

$$AVV^T = U\Sigma V^T.$$

Therefore:

$$\boxed{A = U\Sigma V^T.}$$

The Three Actions

The formula

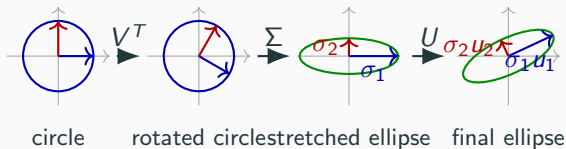
$$A = \underbrace{U}_{\text{output axes}} \underbrace{\Sigma}_{\text{axis lengths}} \underbrace{V^T}_{\text{input coordinates}}$$

means:



Measure, stretch, rebuild.

Circle, Rotate, Stretch, Rotate



The formula is visual: circle $\rightarrow$ circle $\rightarrow$ ellipse $\rightarrow$ rotated ellipse.

Finding the Input Axes

Where Do the Input Axes Come From?

We need directions v_i such that:

$$Av_i = \sigma_i u_i.$$

But A may be rectangular:

$$A : \mathbb{R}^n \longrightarrow \mathbb{R}^m.$$

So A itself may not have eigenvectors.

Instead, measure how much A stretches each input direction.

The Stretch-Measuring Matrix

For any input vector x :

$$\|Ax\|^2 = (Ax)^T (Ax).$$

Move the transpose:

$$(Ax)^T = x^T A^T.$$

Therefore:

$$\|Ax\|^2 = x^T A^T A x.$$

So define:

$$G = A^T A.$$

Why $G = A^T A$ Is Symmetric

Compute:

$$G^T = (A^T A)^T.$$

Use the transpose product rule:

$$(A^T A)^T = A^T (A^T)^T.$$

Since $(A^T)^T = A$:

$$G^T = A^T A = G.$$

$A^T A$ is a real symmetric matrix.

The Squared Length Is Nonnegative

For every vector x :

$$x^T G x = x^T A^T A x.$$

But:

$$x^T A^T A x = (Ax)^T (Ax).$$

So:

$$x^T G x = \|Ax\|^2 \geq 0.$$

We only need this fact: $x^T A^T A x$ is a squared length.

Now Use the Previous Lecture

We proved:

Theorem (Real Symmetric Spectral Theorem)

If G is real symmetric, then it has an orthonormal eigenbasis.

Equivalently,

$$G = V\Lambda V^T$$

for an orthogonal matrix V and a real diagonal matrix Λ .

Here:

$$G = A^T A.$$

Apply It to $A^T A$

Since $A^T A$ is real symmetric:

$$A^T A v_i = \lambda_i v_i$$

with orthonormal eigenvectors:

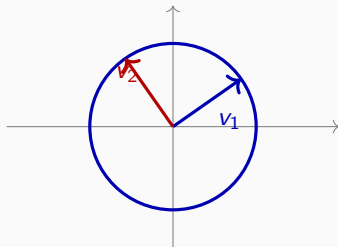
$$v_i^T v_j = \delta_{ij}.$$

Stack them:

$$V = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix}, \quad V^T V = I.$$

These eigenvectors are the input axes of the ellipsoid.

Input Axes Still Live on the Circle



input unit circle

v_i are not output axes yet. They are special directions on the input circle.

Why Eigenvalues Are Nonnegative

Suppose:

$$A^T A v_i = \lambda_i v_i, \quad \|v_i\| = 1.$$

Then:

$$\lambda_i = \lambda_i v_i^T v_i = v_i^T (A^T A v_i) = v_i^T A^T A v_i.$$

But:

$$v_i^T A^T A v_i = \|A v_i\|^2 \geq 0.$$

So:

$$\boxed{\lambda_i \geq 0.}$$

Singular Values Are Square Roots

The eigenvalues of $A^T A$ are nonnegative:

$$\lambda_i \geq 0.$$

So we may define:

$$\sigma_i = \sqrt{\lambda_i}.$$

Then:

$$A^T A v_i = \sigma_i^2 v_i.$$

The square root appears because $A^T A$ measures squared length.

Check the Length

Assume $A^T A v_i = \sigma_i^2 v_i$ and $\|v_i\| = 1$.

Then:

$$\begin{aligned}\|A v_i\|^2 &= v_i^T A^T A v_i \\ &= v_i^T (\sigma_i^2 v_i) \\ &= \sigma_i^2 v_i^T v_i = \sigma_i^2.\end{aligned}$$

Therefore:

$$\|A v_i\| = \sigma_i.$$

Output Axes

Now Build the Output Axes

We already know:

$$A^T A v_i = \sigma_i^2 v_i, \quad \|A v_i\| = \sigma_i.$$

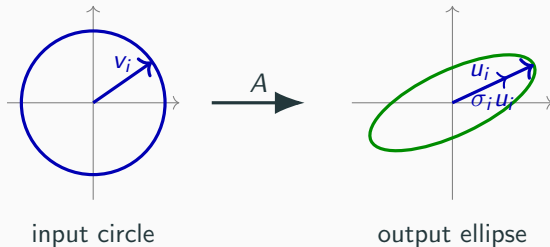
So if $\sigma_i > 0$, define:

$$u_i = \frac{A v_i}{\sigma_i}.$$

Then the axis equation becomes:

$$A v_i = \sigma_i u_i.$$

Input Axis Lands on Output Axis



v_i is an input direction. u_i is the normalized output direction.

Check u_i Has Length 1

By definition:

$$u_i = \frac{Av_i}{\sigma_i}.$$

So:

$$\|u_i\| = \left\| \frac{Av_i}{\sigma_i} \right\| = \frac{\|Av_i\|}{\sigma_i} = \frac{\sigma_i}{\sigma_i} = 1.$$

Each nonzero output axis direction is a unit vector.

Why Different Output Axes Are Orthogonal

For $i \neq j$:

$$u_i^T u_j = \frac{(Av_i)^T (Av_j)}{\sigma_i \sigma_j}.$$

Move A to the middle:

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j}.$$

Use $A^T A v_j = \sigma_j^2 v_j$:

$$u_i^T u_j = \frac{\sigma_j^2}{\sigma_i \sigma_j} v_i^T v_j = 0.$$

Inner-Product Table

The output vectors are orthonormal:

	u_1	u_2	u_3
u_1^T	1	0	0
u_2^T	0	1	0
u_3^T	0	0	1

Matrix form:

$$U^T U = I.$$

What If $\sigma_i = 0$?

If $\sigma_i = 0$, then:

$$A^T A v_i = 0.$$

Compute the output length:

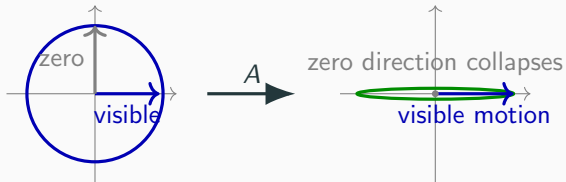
$$\|A v_i\|^2 = v_i^T A^T A v_i = 0.$$

Therefore:

$$A v_i = 0.$$

Zero singular value = invisible input direction.

Visible and Invisible Directions



The null space is exactly the collection of invisible directions.

Complete the Output Basis

The vectors coming from nonzero singular values span $\text{Col}(A)$.

If the output space is larger, complete them:

$$u_1, \dots, u_r \rightsquigarrow u_1, \dots, u_r, u_{r+1}, \dots, u_m.$$

The extra vectors live perpendicular to the column space:

$$\text{Col}(A)^\perp = \text{Null}(A^T).$$

This gives a full orthogonal matrix U .

Callback: Complete by Projection Complement

Collect the existing orthonormal vectors:

$$B = \begin{pmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & \cdots & | \end{pmatrix}.$$

Then the orthogonal projection onto their span is:

$$P = BB^T.$$

The complement projection is:

$$I - P.$$

This is exactly the previous projection-complement method.

Diagonal Cross-Fill the Complement

From previous lectures:

Cross-filling an orthogonal projection gives orthonormal vectors.

Apply that to the complement projection:

$$I - P.$$

Now the missing directions come from the same old machine.

Complement Gives Missing Axes

Cross-filling gives rank-one orthogonal projections:

$$I - P = w_{r+1}w_{r+1}^T + \cdots + w_mw_m^T.$$

Use these vectors as the missing output axes:

$$u_{r+1} = w_{r+1}, \quad \dots, \quad u_m = w_m.$$

Stack Everything

For every input axis:

$$Av_i = \sigma_i u_i.$$

Stack columns:

$$A \underbrace{\begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_V = \underbrace{\begin{pmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_U \underbrace{\Sigma}_{\text{rectangular diagonal}}.$$

So:

$$\boxed{AV = U\Sigma.}$$

Singular Value Decomposition

Since V is orthogonal:

$$V^{-1} = V^T.$$

From:

$$AV = U\Sigma,$$

multiply by V^T :

$$AVV^T = U\Sigma V^T.$$

Therefore:

$$A = U\Sigma V^T.$$

Theorem (Singular Value Decomposition)

For every real $m \times n$ matrix A , there exist orthogonal matrices U and V , and a rectangular diagonal matrix Σ with nonnegative entries, such that

$$A = U\Sigma V^T.$$

Every real matrix maps balls to ellipsoids with orthogonal axes.

Proof Photograph

$A^T A$	is real symmetric	previous lecture applies
$A^T A v_i = \sigma_i^2 v_i$	gives	input axes v_i
$\ A v_i\ = \sigma_i$	gives	axis lengths σ_i
$u_i = A v_i / \sigma_i$	gives	output axes u_i
$A v_i = \sigma_i u_i$	stack	$AV = U\Sigma$
$V^{-1} = V^T$	gives	$A = U\Sigma V^T$

The whole proof is real symmetric diagonalization of $A^T A$.

Method Summary

The Method

To decompose a real matrix A :

1. Compute the real symmetric matrix $A^T A$.
2. Orthogonally diagonalize it:

$$A^T A = V \Sigma^2 V^T.$$

3. For each nonzero σ_i , define:

$$u_i = \frac{A v_i}{\sigma_i}.$$

4. Complete by the previous projection method:

$$B = (u_1 \ \cdots \ u_r), \quad P = B B^T, \quad I - P.$$

Diagonal cross-fill $I - P$.

5. Stack the equations $A v_i = \sigma_i u_i$.

What Each Piece Means

The decomposition is:

$$A = U\Sigma V^T.$$

Piece	Meaning
V	orthonormal input axes on the unit circle
Σ	axis lengths of the output ellipsoid
U	orthonormal output axes

Input axes $\rightarrow$ stretch lengths $\rightarrow$ output axes.

The Whole Story in One Line

For every input axis:

$$Av_i = \sigma_i u_i.$$

Stack all columns:

$$A \underbrace{\begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_V = \underbrace{\begin{pmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_U \underbrace{\Sigma}_{\text{axis lengths}}.$$

Therefore:

$$A = U\Sigma V^T.$$

SVD answers one geometric question:

What does A do to a round ball?

Answer:

round ball $\xrightarrow{A}$ ellipsoid with orthogonal axes.

The axis data is exactly:

$$A = U\Sigma V^T.$$